Lecture 9 (10/5)

Bias, consistency, and the bias-variance trade-off

Bias and consistency

For any estimator, we define the mean squared error

$$\mathrm{MSE}_{\hat{\theta}} = E\left[(\hat{\theta} - \theta)^2\right]$$

Bias and consistency

For any estimator, we define the mean squared error

$$\mathrm{MSE}_{\hat{\theta}} = E\left[(\hat{\theta} - \theta)^2\right]$$

and we have the Bias-variance decomposition

$$\mathrm{MSE}_{\hat{\theta}} = \mathrm{se}(\hat{\theta})^2 + \left(E[\hat{\theta}] - \theta\right)^2.$$

Bias and consistency

For any estimator, we define the mean squared error

$$\mathrm{MSE}_{\hat{\theta}} = E\left[(\hat{\theta} - \theta)^2\right]$$

and we have the Bias-variance decomposition

$$\mathrm{MSE}_{\hat{\theta}} = \mathrm{se}(\hat{\theta})^2 + \left(E[\hat{\theta}] - \theta\right)^2.$$

The bias is defined as $\mathrm{Bias} = E[\hat{\theta}] - \theta$ and $\hat{\theta}$ is unbiased if the bias is zero.

Bias and consistency

For any estimator, we define the mean squared error

$$\mathrm{MSE}_{\hat{\theta}} = E\left[(\hat{\theta} - \theta)^2\right]$$

and we have the Bias-variance decomposition

$$\mathrm{MSE}_{\hat{\theta}} = \mathrm{se}(\hat{\theta})^2 + \left(E[\hat{\theta}] - \theta\right)^2.$$

The bias is defined as $\mathrm{Bias} = E[\hat{\theta}] - \theta$ and $\hat{\theta}$ is unbiased if the bias is zero. $\hat{\theta}$ is consistent if

$$E[\hat{\theta}] \to \theta \quad \text{and} \quad \mathrm{se}(\hat{\theta}) \to 0 \quad \text{as } N \to \infty.$$

Bias and consistency

For any estimator, we define the mean squared error

$$\text{MSE}_{\hat{\theta}} = E\left[(\hat{\theta} - \theta)^2\right]$$

and we have the Bias-variance decomposition

$$\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + \left(E[\hat{\theta}] - \theta\right)^2.$$

The bias is defined as $\text{Bias} = E[\hat{\theta}] - \theta$ and $\hat{\theta}$ is unbiased if the bias is zero. $\hat{\theta}$ is consistent if

$$E[\hat{\theta}] \to \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \to 0 \quad \text{as } N \to \infty.$$

Question

Last class we looked at the sample mean $\overline{Y}$ as an estimator for both Bernoulli and Normal data. Is it unbiased? Is it consistent?

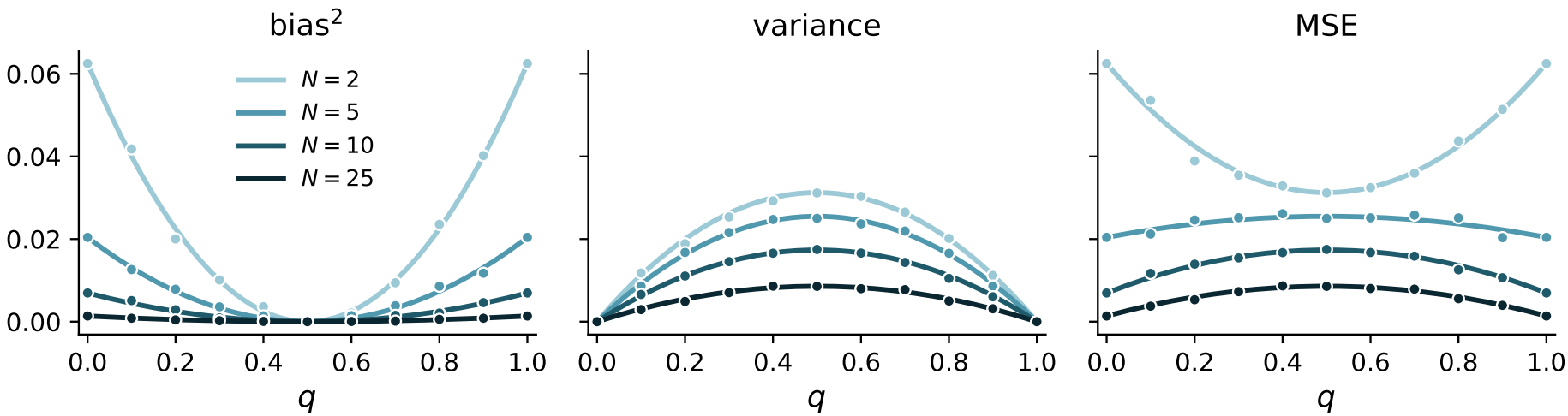
Question

Let $Y_1, \ldots, Y_N \sim \text{Bernoulli}(q)$. The natural estimator of q is $\hat{q} = \overline{Y}$. However, if every $Y_i = 0$, then $\hat{q} = 0$. This is problematic: if we flip a coin twice and get two tails, we would conclude that the coin can never land on heads. To correct for this, we pretend we have already seen two flips, one heads and one tails. Laplace's rule of succession is the estimator

$$\hat{q}_L = \frac{S+1}{N+2}, \qquad S = \sum_i Y_i.$$

Find the bias, variance and MSE of $\hat{q}_L$.

$$\text{bias} = \frac{1-2q}{N+2}, \qquad \text{se}(\hat{q}_L)^2 = \frac{Nq(1-q)}{(N+2)^2}$$



Curves are formulas above. Points are estimates from 1000 simulated replicates (see code on next slide).

```
N = 10
R = 1000  # number of replicates
qs = np.linspace(0, 1, 11)
for i in range(len(qs)):
    q = qs[i]
    q_hat = np.zeros(R)
    for r in range(R):
        S = np.random.binomial(N, q)
        # This also works:
        # ys = np.random.choice([0, 1], N, p=[1-q, q])
        # S = np.sum(ys)
        q_hat[r] = (S + 1) / (N + 2)
    bias2[i] = (np.mean(q_hat) - q)**2
    var[i] = np.var(q_hat)
    mse[i] = np.mean((q_hat - q)**2)
```

Each pass through the inner loop is one replicate of the data, so `q_hat` holds R draws from the sample distribution of $\hat{q}_L$. We repeat this for $N = 2, 5, 10, 25$.

Question

Suppose $Y_i \sim \mathrm{Normal}(\mu, \sigma^2)$ and μ is known to us

$$\hat{\sigma}^2_\mu = \frac{1}{N} \sum_i (Y_i - \mu)^2$$

Is this an unbiased estimator?

Question

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is known to us

$$\hat{\sigma}^2_\mu = \frac{1}{N}\sum_i (Y_i - \mu)^2$$

Is this an unbiased estimator?

Question

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is NOT known to us

$$\hat{\sigma}^2_0 = \frac{1}{N}\sum_i (Y_i - \bar{Y})^2$$

Is this an unbiased estimator?

Bessel's correction

Why is $\hat{\sigma}_0^2 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$ biased? Add and subtract μ, expand, and use $\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)$:

$$\sum_i (Y_i - \overline{Y})^2 = \sum_i \left[(Y_i - \mu) - (\overline{Y} - \mu)\right]^2$$

Bessel's correction

Why is $\hat{\sigma}_0^2 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$ biased? Add and subtract μ, expand, and use $\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)$:

$$\begin{aligned}\sum_i (Y_i - \overline{Y})^2 &= \sum_i \left[(Y_i - \mu) - (\overline{Y} - \mu)\right]^2 \\ &= \sum_i (Y_i - \mu)^2 - 2(\overline{Y} - \mu)\sum_i (Y_i - \mu) + N(\overline{Y} - \mu)^2\end{aligned}$$

Bessel's correction

Why is $\hat{\sigma}_0^2 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$ biased? Add and subtract μ, expand, and use $\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)$:

$$\begin{aligned}
\sum_i (Y_i - \overline{Y})^2 &= \sum_i \left[(Y_i - \mu) - (\overline{Y} - \mu)\right]^2 \\
&= \sum_i (Y_i - \mu)^2 - 2(\overline{Y} - \mu)\sum_i (Y_i - \mu) + N(\overline{Y} - \mu)^2 \\
&= \sum_i (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2
\end{aligned}$$

Bessel's correction

Why is $\hat{\sigma}_0^2 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$ biased? Add and subtract μ, expand, and use $\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)$:

$$\begin{aligned}
\sum_i (Y_i - \overline{Y})^2 &= \sum_i \left[(Y_i - \mu) - (\overline{Y} - \mu)\right]^2 \\
&= \sum_i (Y_i - \mu)^2 - 2(\overline{Y} - \mu)\sum_i (Y_i - \mu) + N(\overline{Y} - \mu)^2 \\
&= \sum_i (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2
\end{aligned}$$

Taking expectations, $E[(Y_i - \mu)^2] = \sigma^2$ and $E[(\overline{Y} - \mu)^2] = \mathsf{var}(\overline{Y}) = \sigma^2/N$, so $E\left[\sum_i (Y_i - \overline{Y})^2\right] = N\sigma^2 - N \cdot \sigma^2/N = (N-1)\sigma^2$.

Bessel's correction

Why is $\hat{\sigma}_0^2 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$ biased? Add and subtract μ, expand, and use $\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)$:

$$\begin{aligned}\sum_i (Y_i - \overline{Y})^2 &= \sum_i \left[(Y_i - \mu) - (\overline{Y} - \mu)\right]^2 \\ &= \sum_i (Y_i - \mu)^2 - 2(\overline{Y} - \mu)\sum_i (Y_i - \mu) + N(\overline{Y} - \mu)^2 \\ &= \sum_i (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2\end{aligned}$$

Taking expectations, $E[(Y_i - \mu)^2] = \sigma^2$ and $E[(\overline{Y} - \mu)^2] = \mathsf{var}(\overline{Y}) = \sigma^2/N$, so $E\left[\sum_i (Y_i - \overline{Y})^2\right] = N\sigma^2 - N \cdot \sigma^2/N = (N-1)\sigma^2$.
So $E[\hat{\sigma}_0^2] = \frac{N-1}{N}\sigma^2$: biased, but consistent. Dividing by $N-1$ instead (Bessel's correction) gives an unbiased estimator

$$\boxed{\hat{\sigma}^2 = \frac{1}{N-1}\sum_i (Y_i - \overline{Y})^2, \qquad E[\hat{\sigma}^2] = \sigma^2}$$

Solutions

1) Sample mean $\overline{Y}$: $E[\overline{Y}] = q$ (Bernoulli) or μ (Normal), so it is unbiased. $\text{se}(\overline{Y}) = \sqrt{q(1-q)/N}$ or $\sigma/\sqrt{N}$, both $\to 0$ as $N \to \infty$, so it is consistent.

2) Laplace's rule, $S \sim \text{Binomial}(N, q)$: $E[\hat{q}_L] = \dfrac{Nq+1}{N+2}$, so bias $= \dfrac{1-2q}{N+2}$; $\text{var}(\hat{q}_L) = \dfrac{Nq(1-q)}{(N+2)^2}$; $\text{MSE} = \dfrac{Nq(1-q)+(1-2q)^2}{(N+2)^2}$. Biased (unless $q = \frac{1}{2}$) but consistent.

3) μ known: $E[(Y_i - \mu)^2] = \sigma^2$ for each i, so $E[\hat{\sigma}^2_\mu] = \sigma^2$. Yes, unbiased.

4) μ not known: $\sum_i (Y_i - \overline{Y})^2 = \sum_i (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2$ and $E[(\overline{Y} - \mu)^2] = \sigma^2/N$, so $E[\hat{\sigma}^2_0] = \dfrac{N-1}{N}\sigma^2$. No, biased: bias $= -\sigma^2/N$ (it underestimates σ^2). The bias $\to 0$, so it is still consistent; dividing by $N-1$ instead of N removes it.